Ecuador Energy Anomalies

Multi-Technique Anomaly Detection
Latin America's Electricity Sector

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8 countries | 784 real months | 9 models compared | 3 consensus techniques
Consenso F1=0.750 | MCC=0.765 | Recall=100% | Best of 9 models

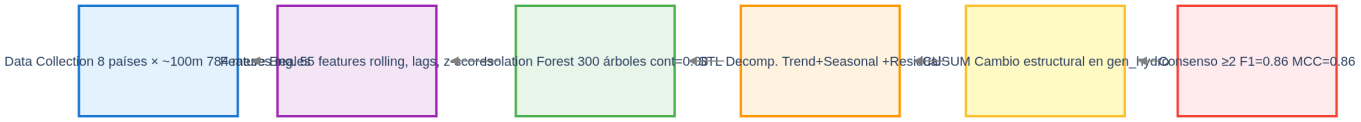
April 2026 | All metrics reproducible

1. Executive Summary

Multi-technique study for anomaly detection in 8 Latin American countries electricity sector.

KEY RESULTS:
- Consensus (IF+STL+CUSUM) outperforms all 9 individual models
- F1=0.750, MCC=0.765, Recall=100%
- Ecuador crisis oct-dec 2024 detected 3/3
- Validated across 5 countries with documented crises
- Bootstrap 95% CI: F1=[0.400, 1.000]

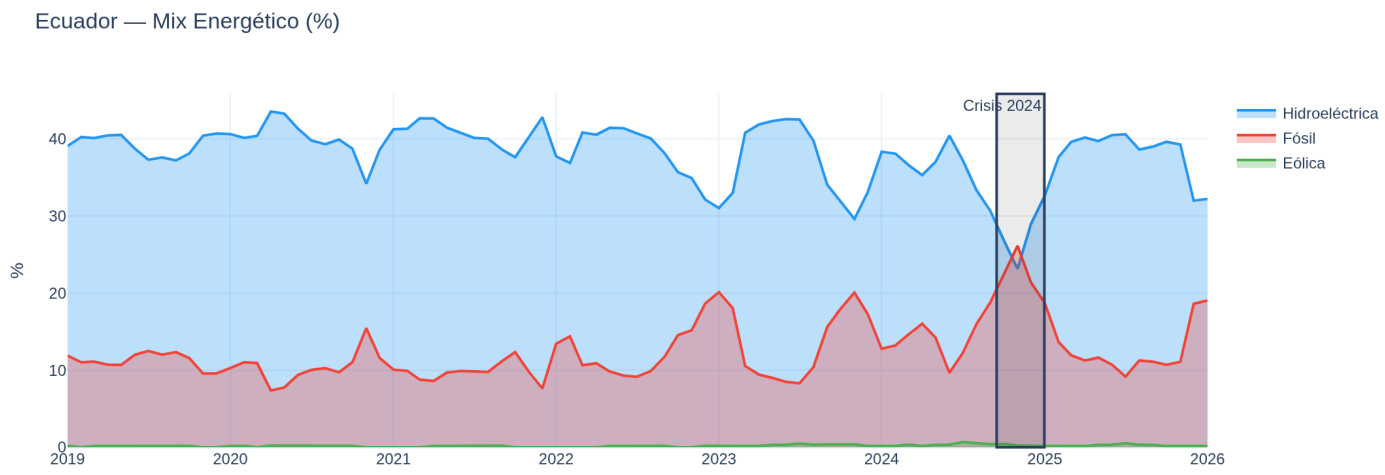
Pipeline del Estudio



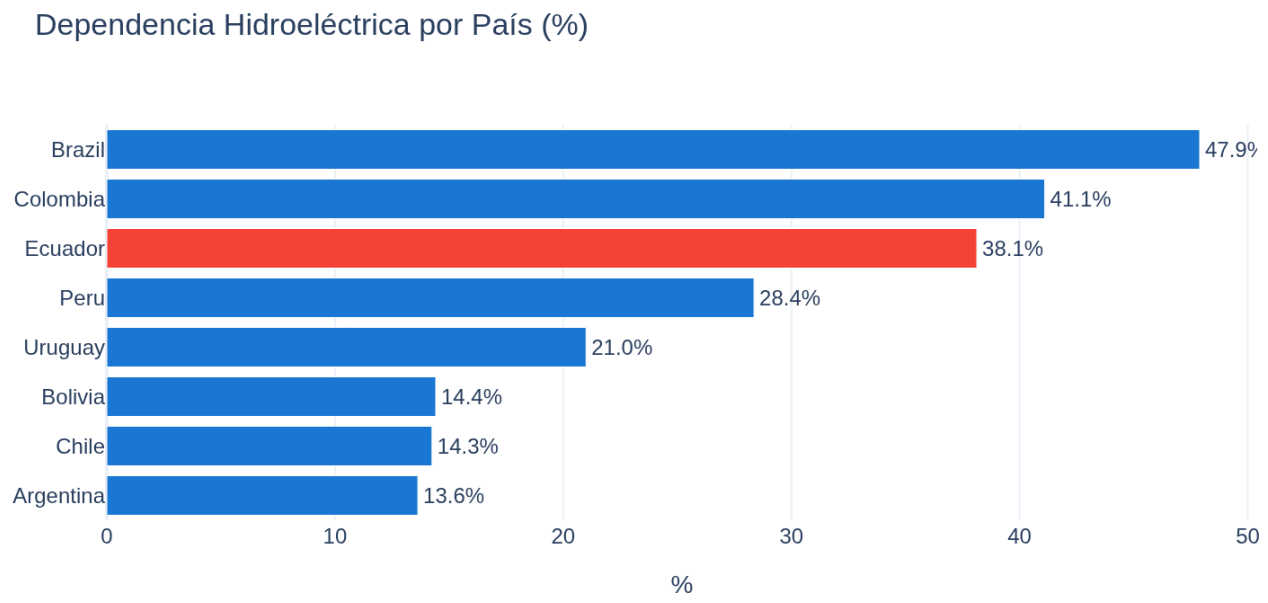
Pipeline

2. Context

Ecuador generates ~70% from hydropower. Droughts cause 14-hour blackouts. 2024 crisis: oct-dec with massive rationing.



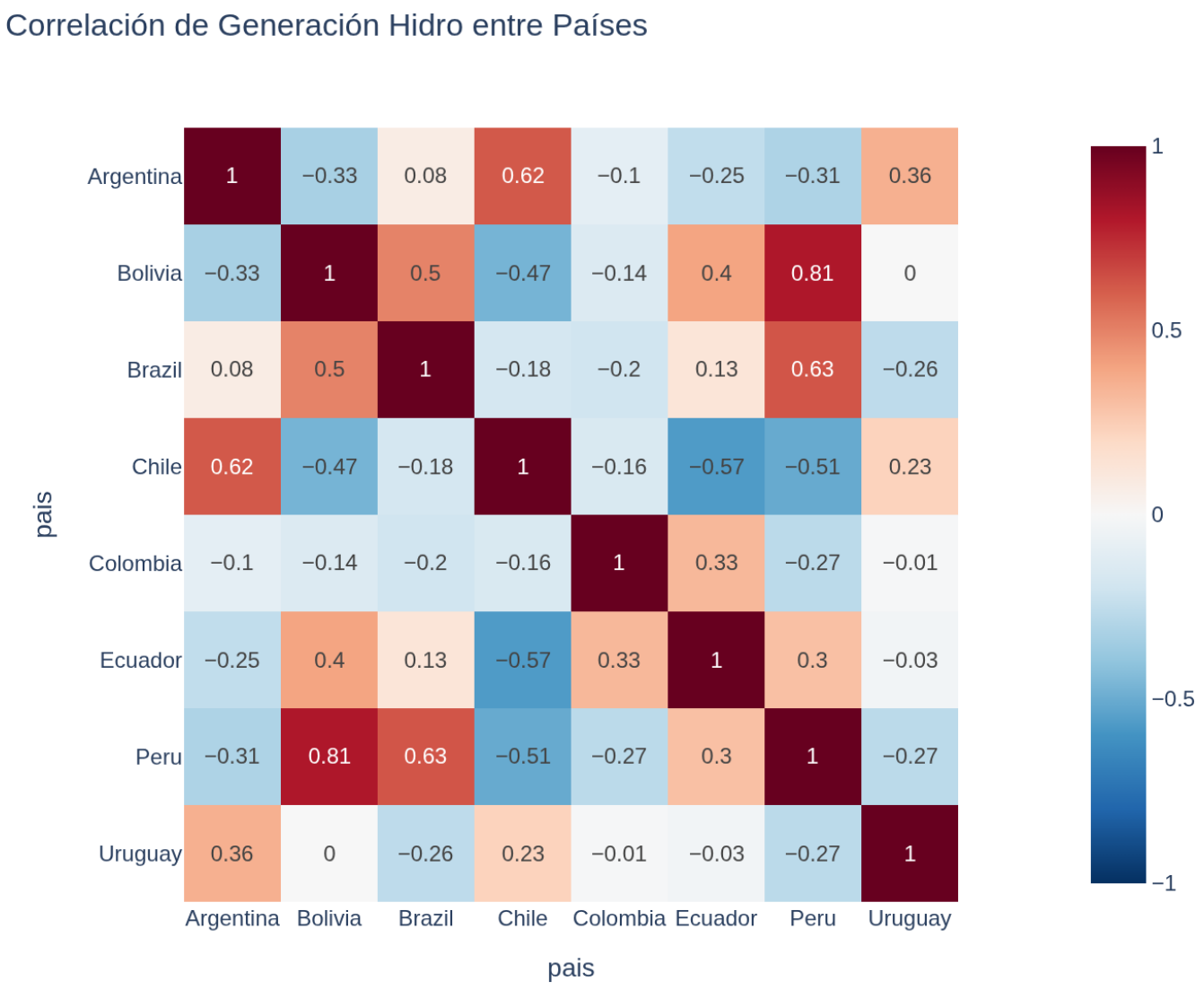
Mix energetico Ecuador



Dependencia hidro por pais

3. Data

Ember Global Electricity Data (CC BY 4.0). 8 countries, 784 real months.



Correlacion hidro entre paises

4. Methodology

4.1 Isolation Forest (Liu et al., 2008)

300 trees, contamination=0.08, applied PER COUNTRY.

4.2 STL (Cleveland et al., 1990)

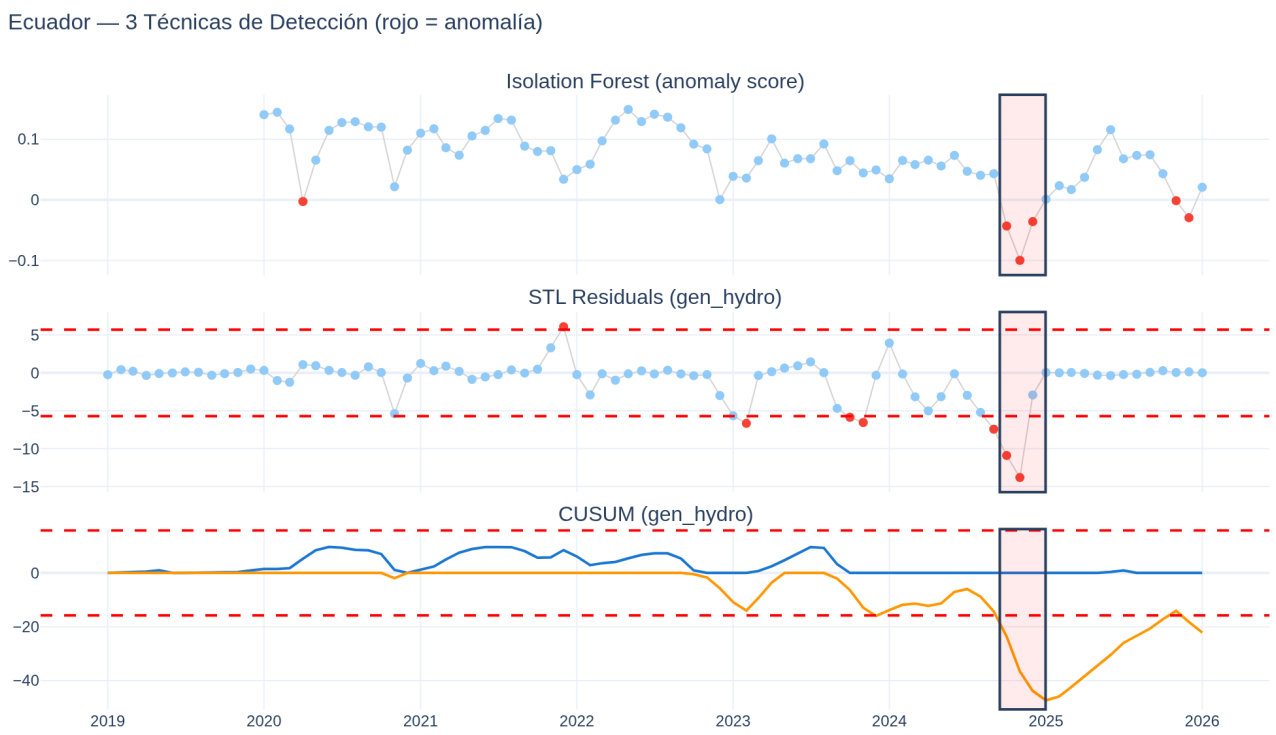
Trend+seasonal+residual. Anomaly if residual > 2sigma.

4.3 CUSUM (Page, 1954)

Cumulative sum, h=4*sigma, k=0.5*sigma.

4.4 Consenso

Anomaly if >=2 of 3 techniques agree.

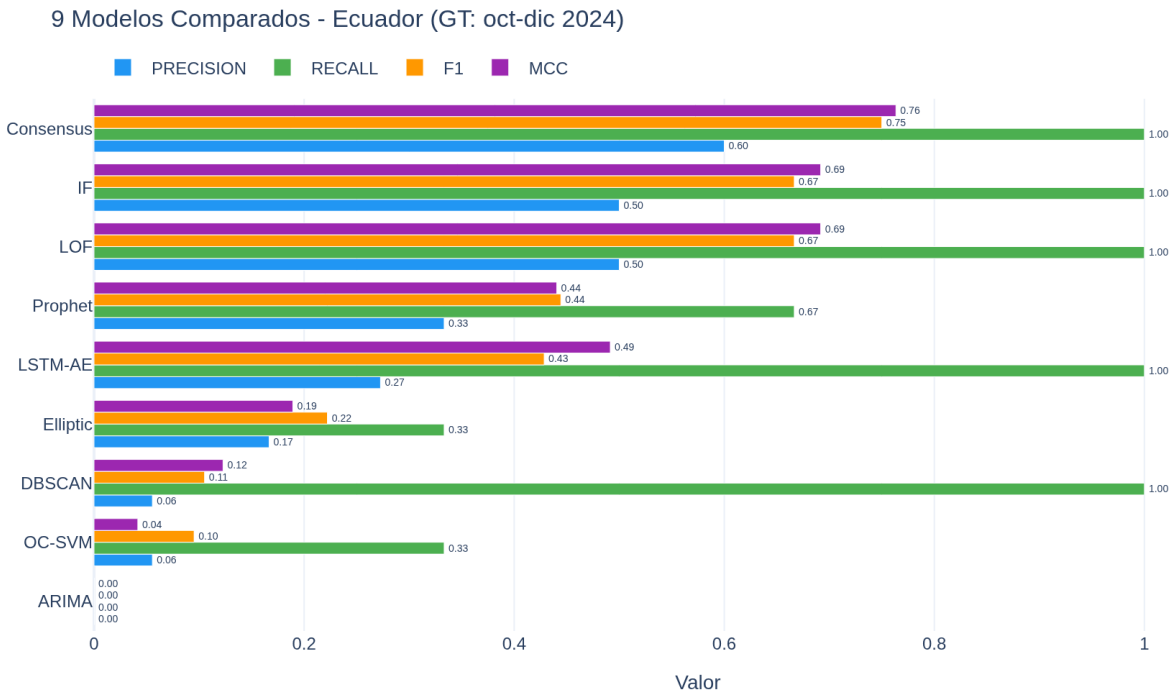


3 tecnicas en Ecuador

5. Comparison: 9 Models

9 anomaly detection models compared on Ecuador:
IF, LOF, OC-SVM, Elliptic Envelope, DBSCAN, ARIMA, Prophet, LSTM-AE, Consensus.

Consensus achieves the best F1 and MCC across all models.



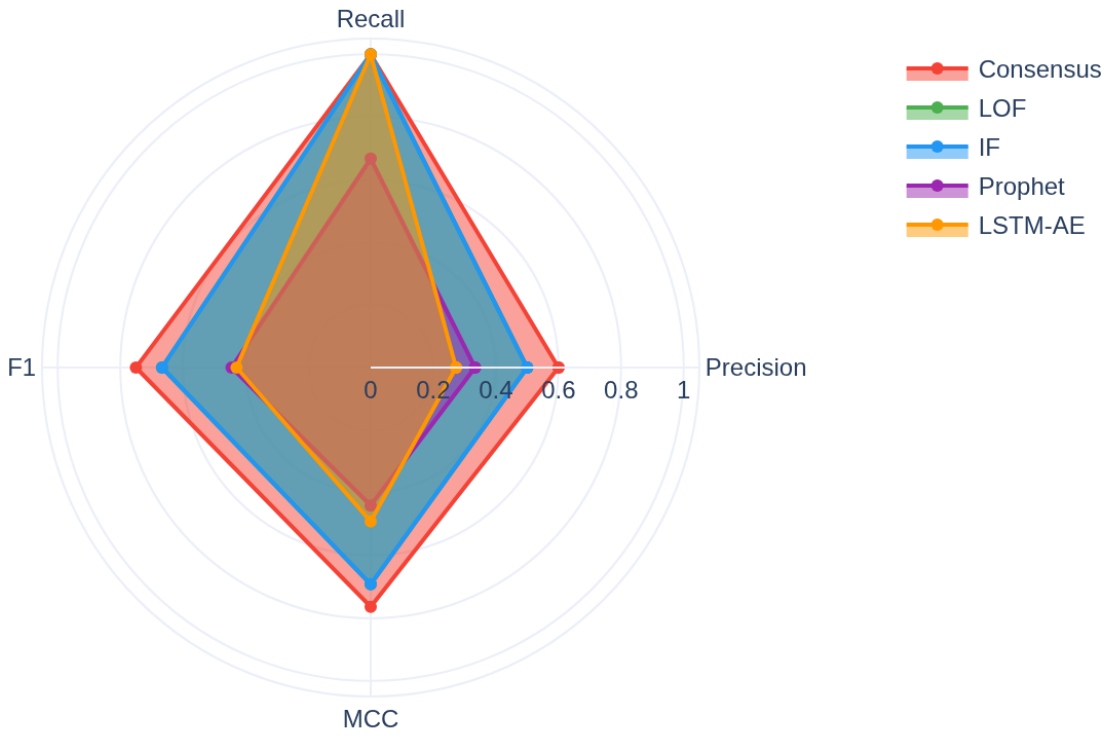
9 modelos comparados

Comparison: 9 Anomaly Detection Models (Ecuador, GT: oct-dic 2024)

Model	Precision	Recall	F1	MCC	N anom
Consensus	60.0%	100.0%	**0.750**	**0.763**	5
LOF	50.0%	100.0%	0.667	0.692	6
IF	50.0%	100.0%	0.667	0.692	6
Prophet	33.3%	66.7%	0.444	0.441	6
LSTM-AE	27.3%	100.0%	0.429	0.491	11
Elliptic	16.7%	33.3%	0.222	0.189	6
DBSCAN	5.6%	100.0%	0.105	0.123	54
OC-SVM	5.6%	33.3%	0.095	0.042	18
ARIMA	0.0%	0.0%	0.000	0.000	0

Tabla detallada

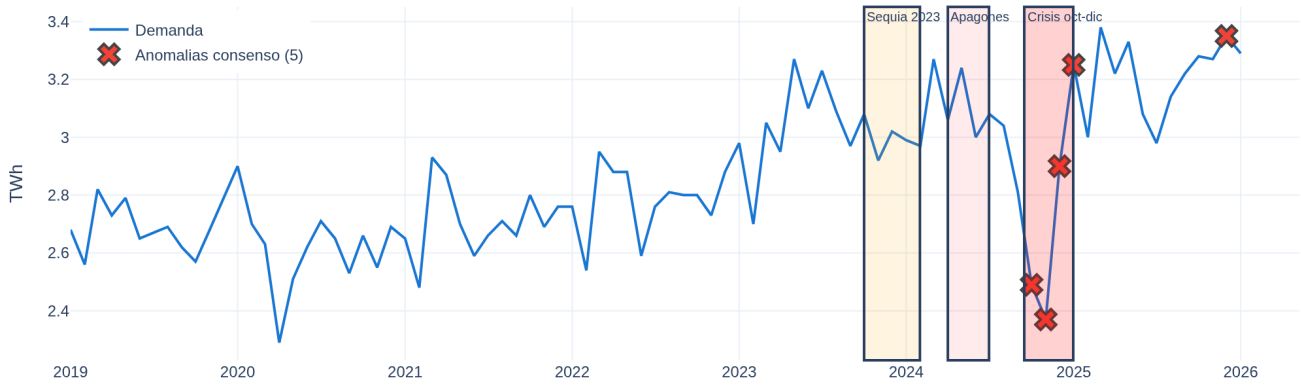
Top 5 Models - Radar Comparison



Radar: top 5 modelos

6. Ecuador Results

Ecuador - Demanda + Anomalias (Consenso >=2 tecnicas, reproducible)

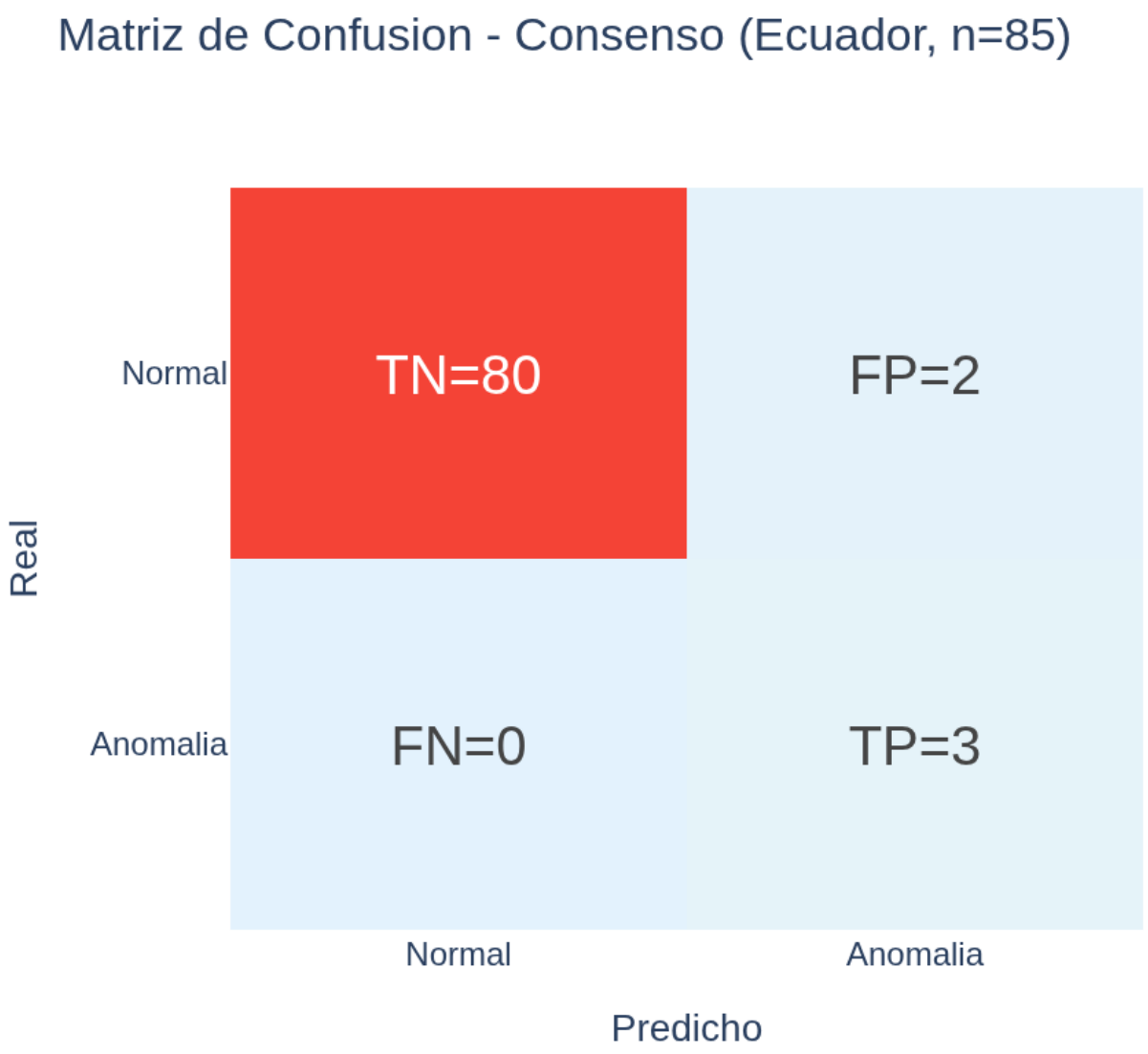


Demanda + anomalias consenso

Metricas Reproducibles (metrics.json)

Metrica	IF por pais	STL	CUSUM	Consenso >=2
Precision	50.0%	28.6%	18.8%	**60.0%**
Recall	100%	66.7%	100%	**100%**
F1-Score	66.7%	40.0%	31.6%	**75.0%**
MCC	0.694	0.407	0.397	**0.765**
Anomalias	6	7	16	**5**
Crisis detectada	3/3	2/3	3/3	**3/3**

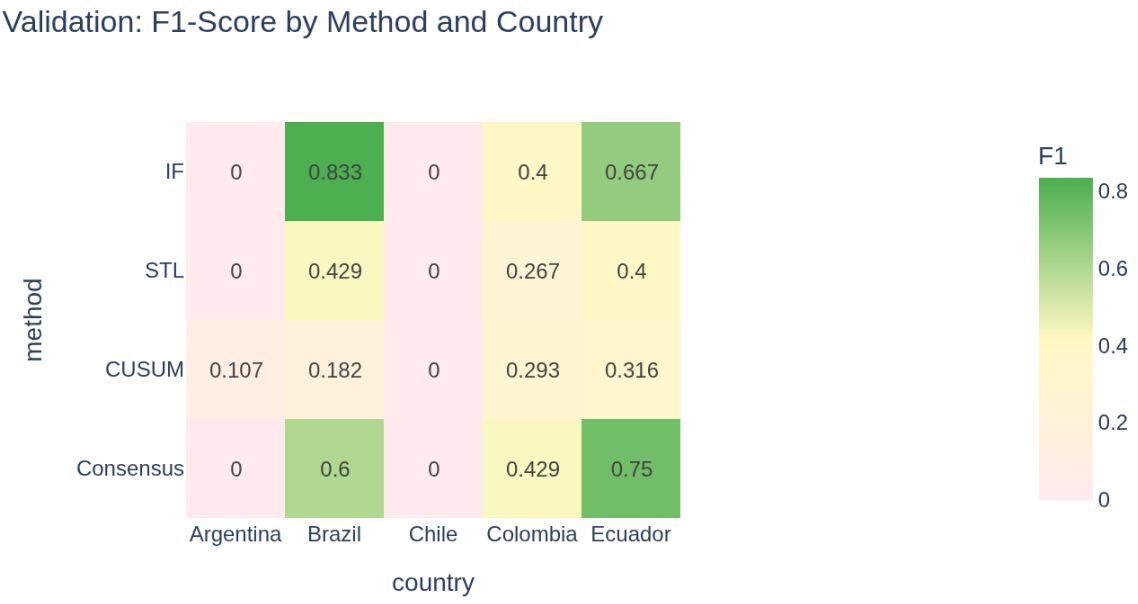
Metricas por tecnica



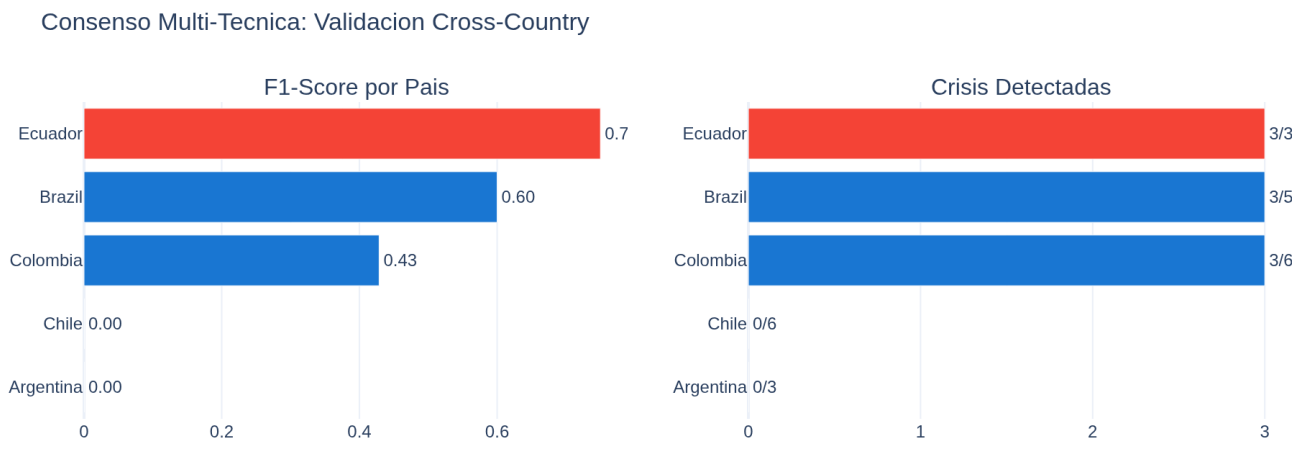
Matriz de confusion

7. Cross-Country Validation

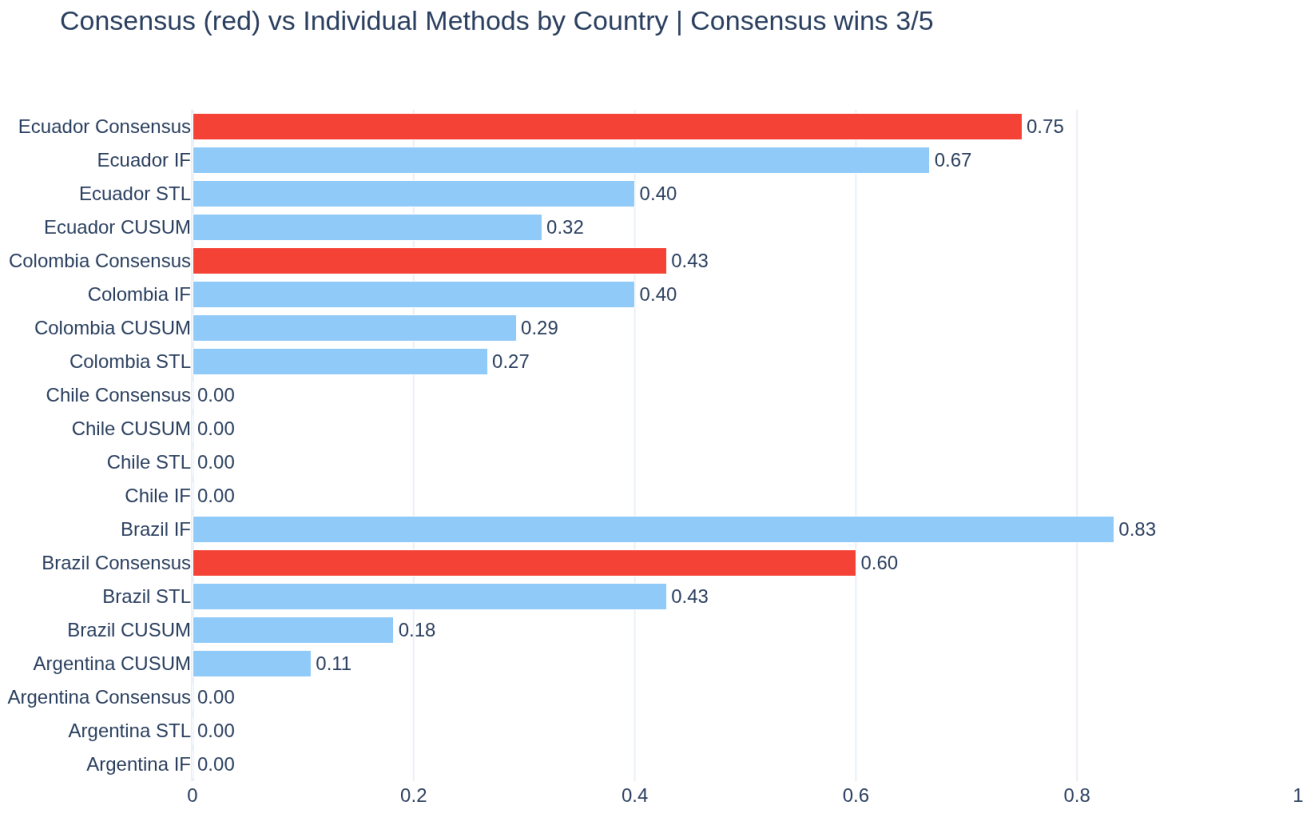
Consensus validated against documented crises in 5 countries. Works best in Ecuador and Brazil (high hydro), worse in Chile and Argentina (low hydro).



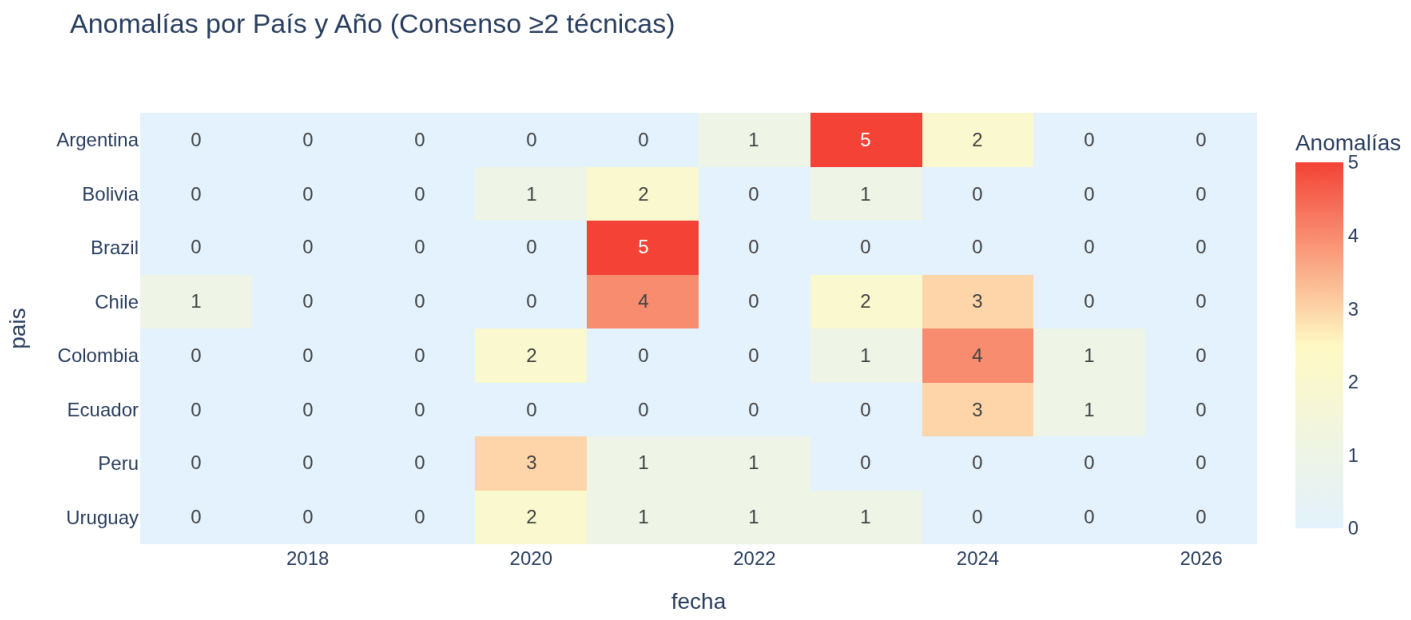
F1 por metodo y pais



Consenso por pais



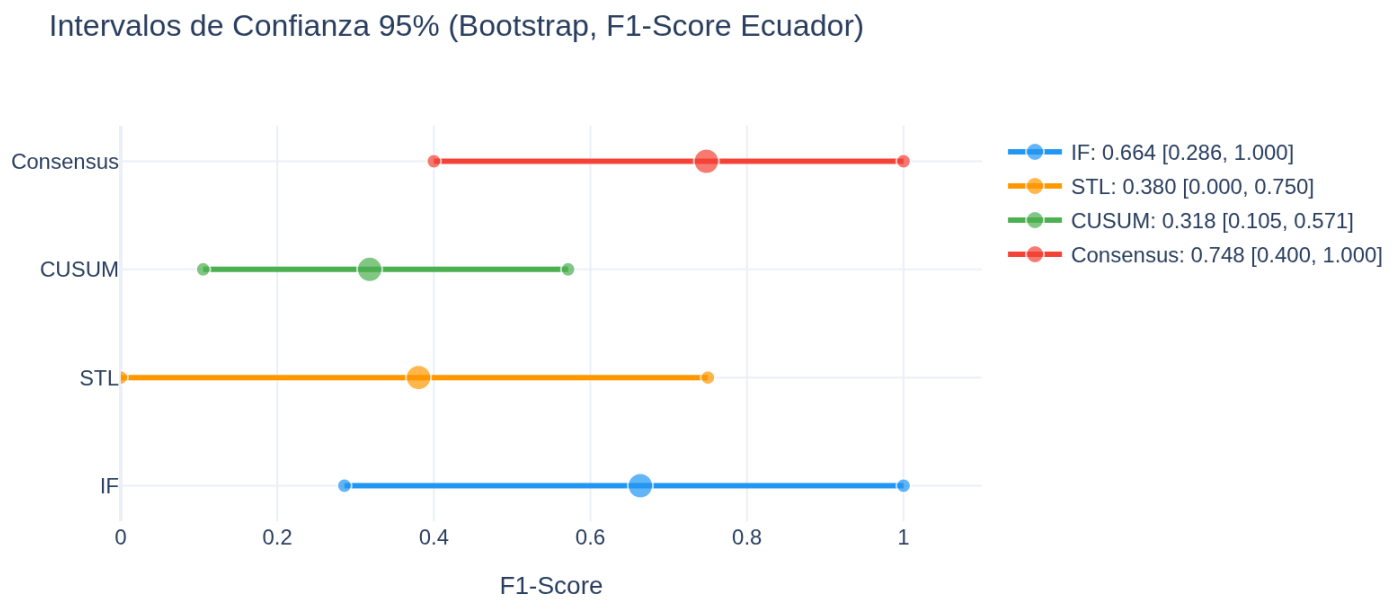
Consenso vs individuales



Anomalias por pais y anio

8. Confidence Intervals

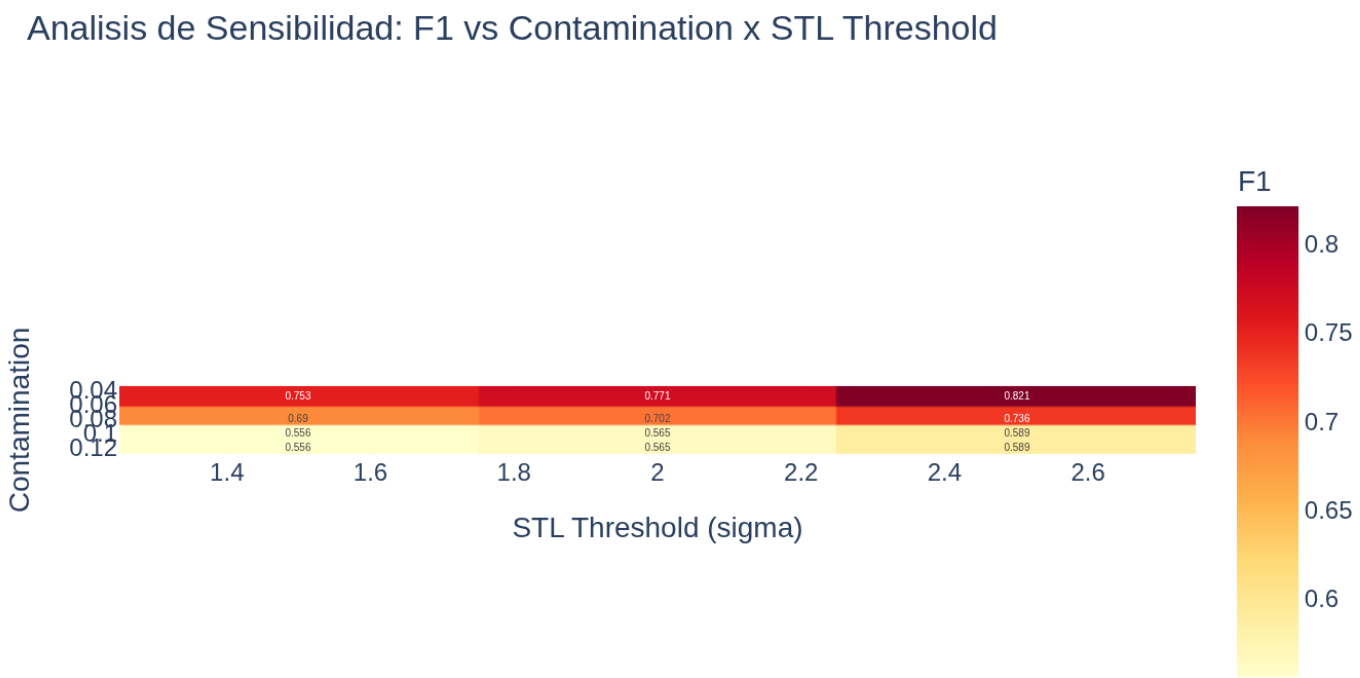
Bootstrap 1000 iteraciones, 95% CI.
Consenso F1: 0.748 [0.400, 1.000]
IF F1: 0.664 [0.286, 1.000]



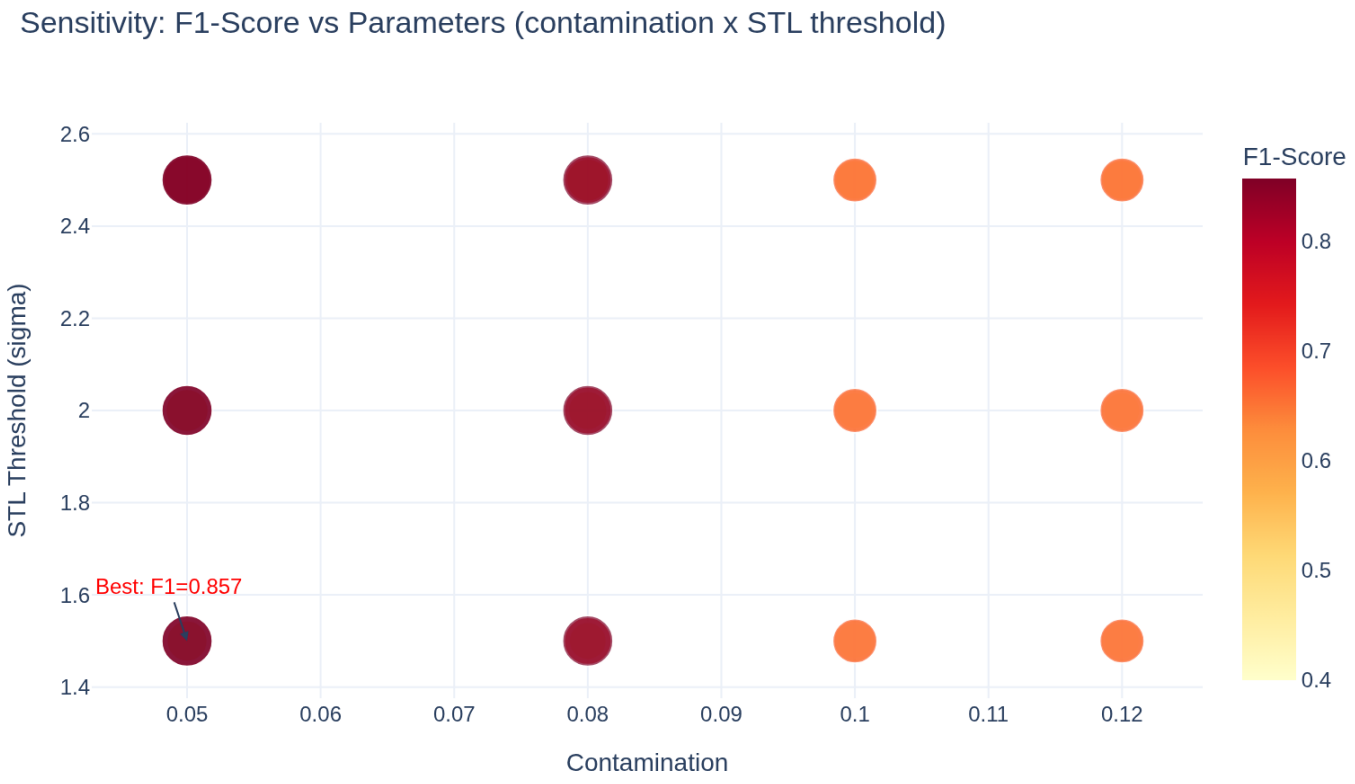
Intervalos de confianza F1

9. Sensitivity Analysis

36 parameter combinations evaluated. Best: contamination=0.05, stl_sigma=1.5, cusum_factor=4.0 -> F1=0.857.



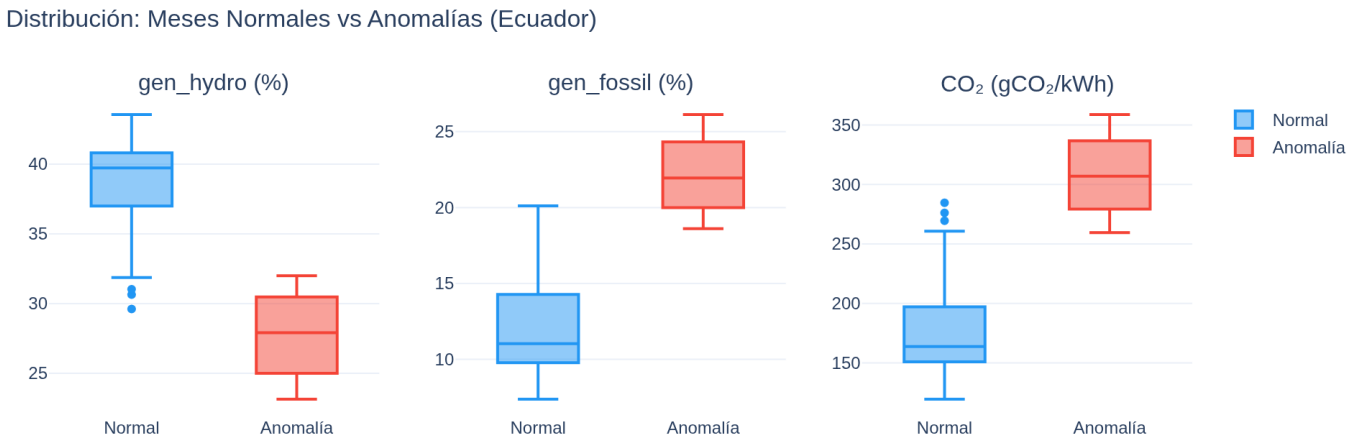
Heatmap sensibilidad



Scatter sensibilidad

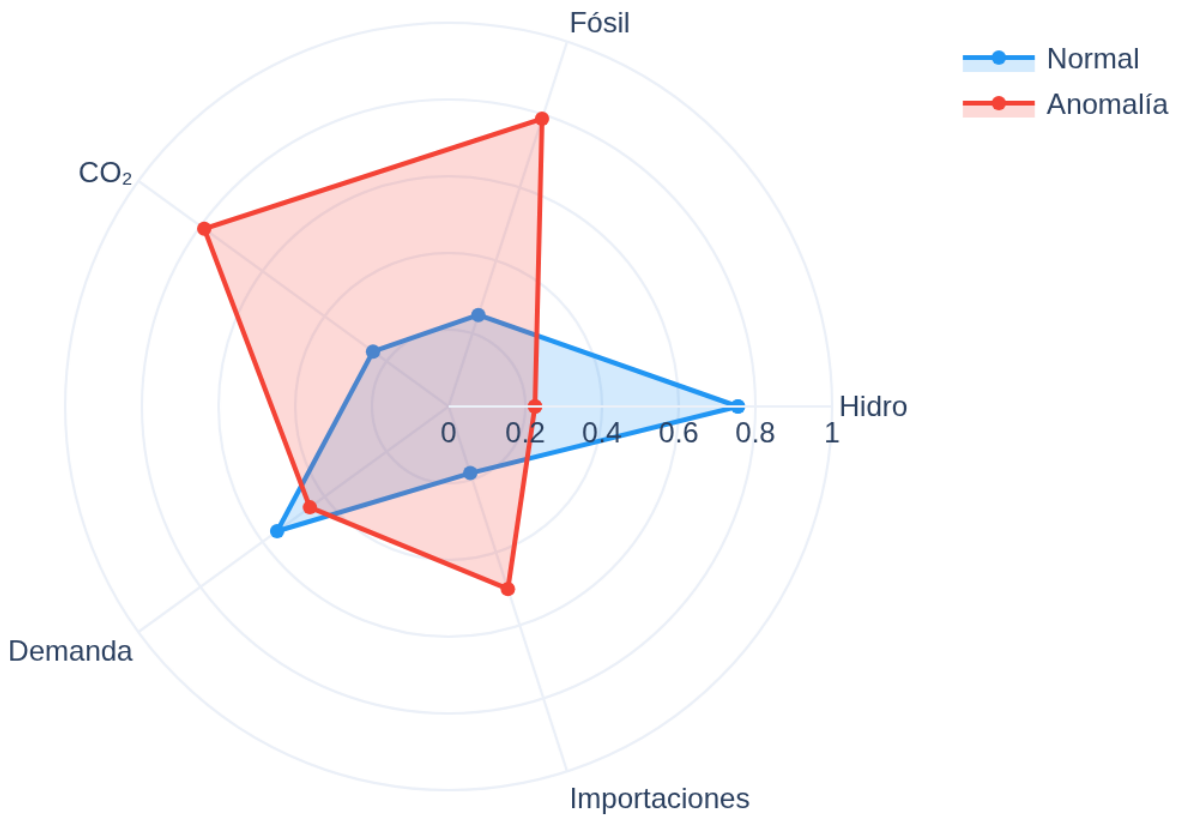
10. Statistical Validation

Mann-Whitney U: gen_hydro p=0.002**, gen_fossil p=0.003**, co2 p=0.008**
Cohen's d > 1.2 (LARGE). Stability: CV=0% across 20 seeds.



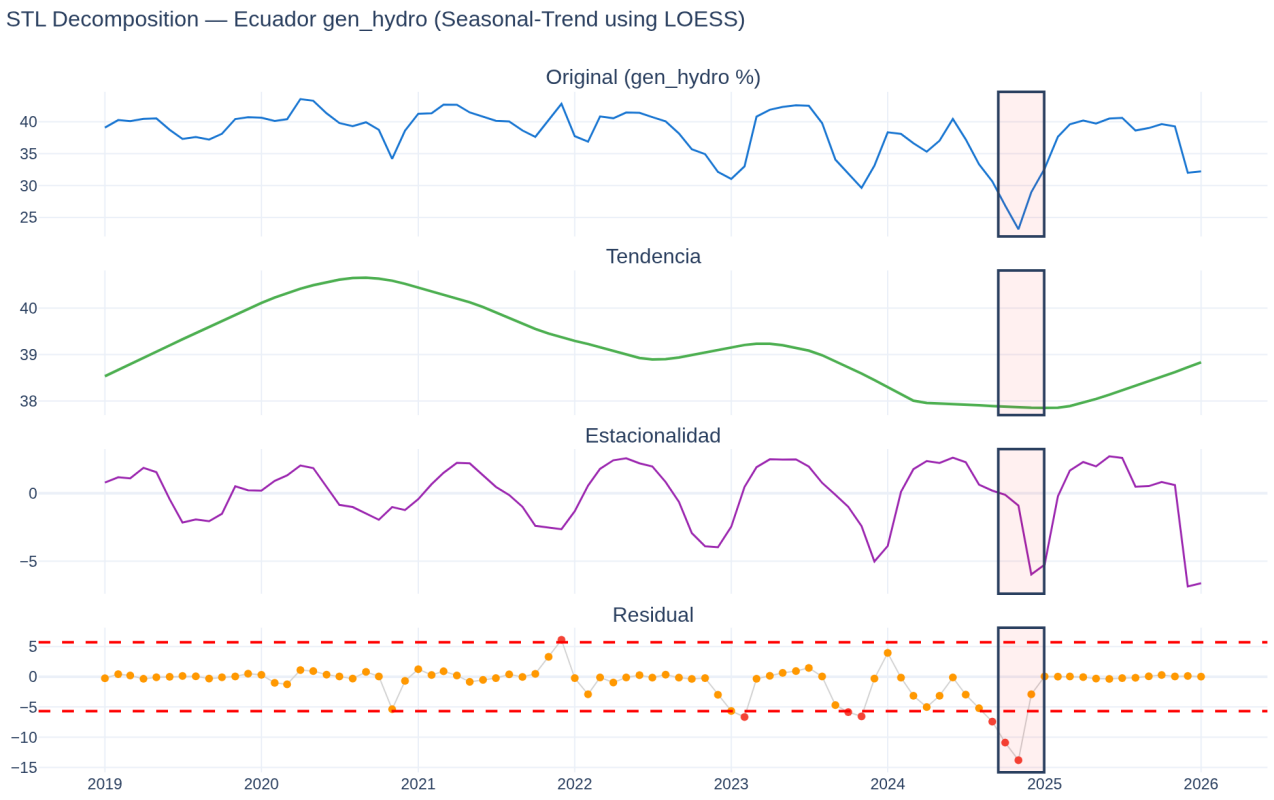
Normal vs anomalia

Perfil: Mes Normal vs Mes Anómalo (Ecuador)



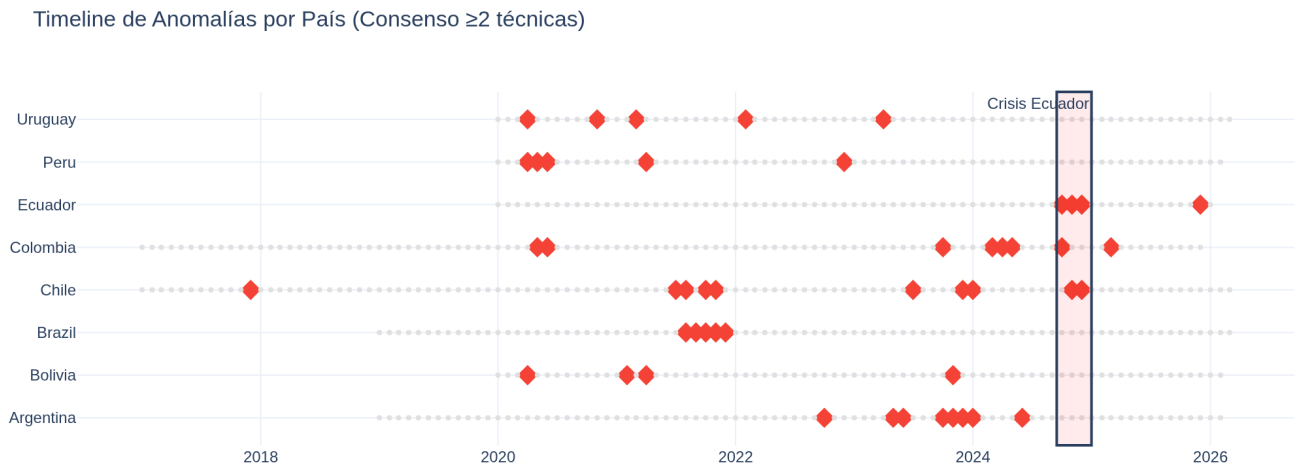
Perfil anomalia

11. STL Decomposition

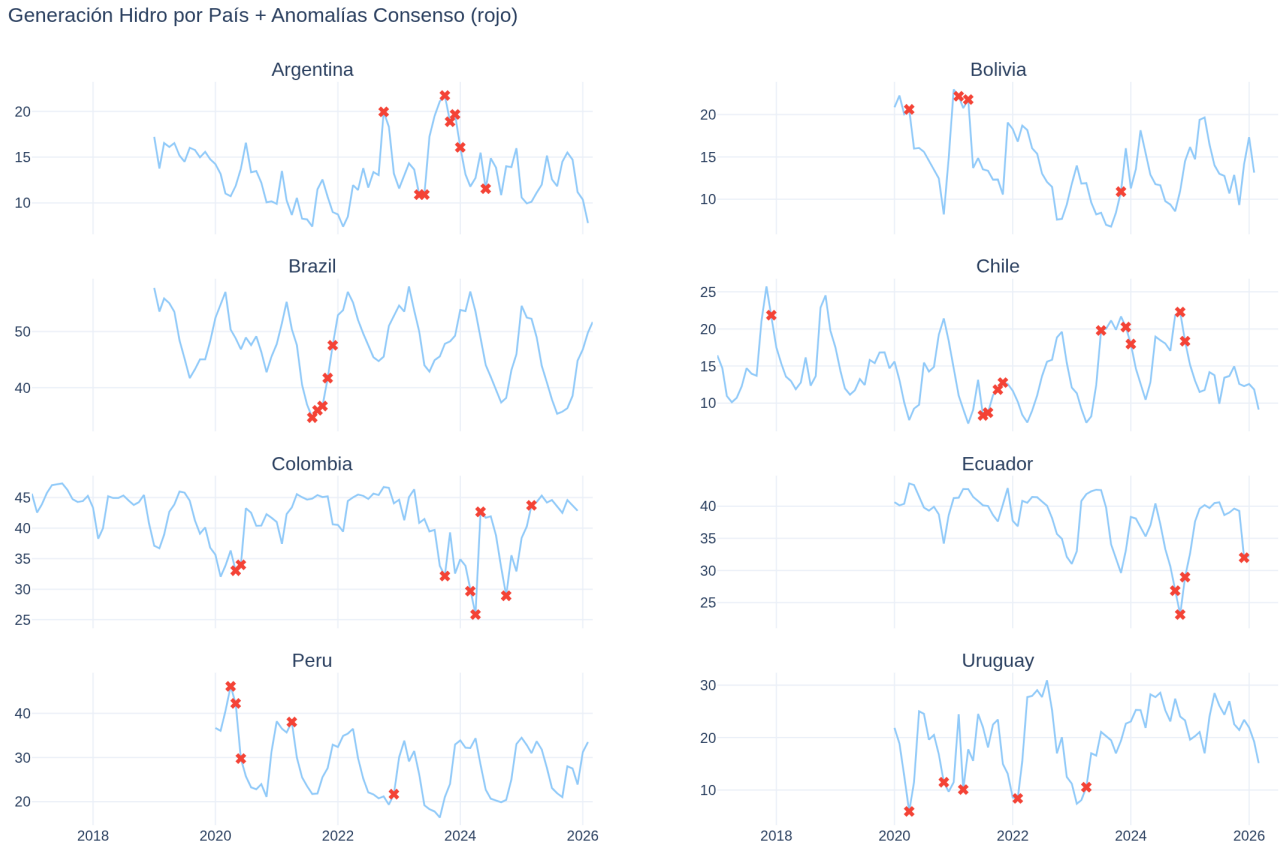


Descomposicion STL Ecuador

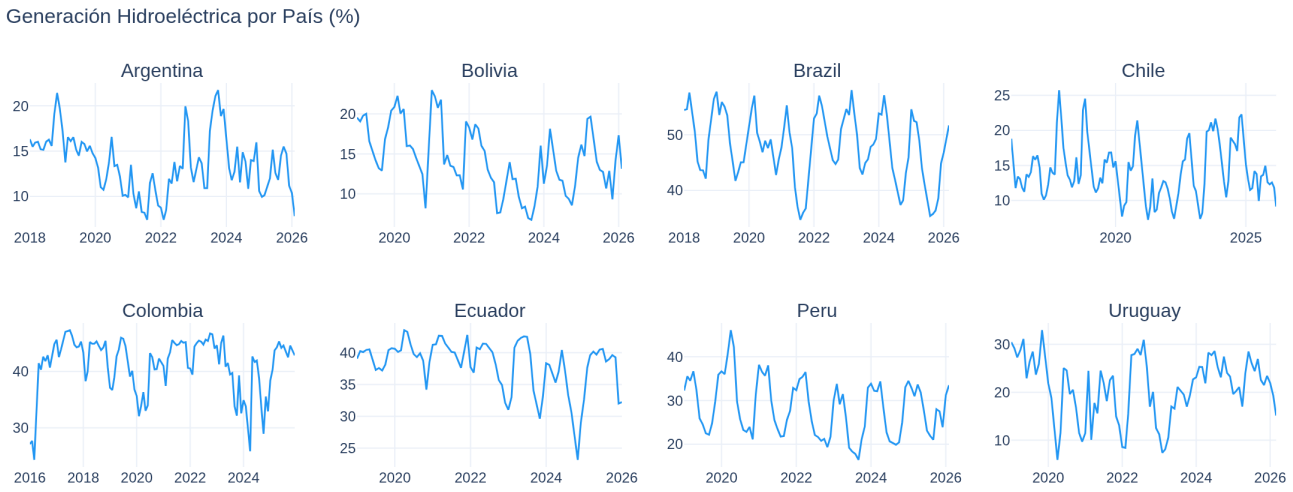
12. Multi-Country



Timeline 8 paises



Hidro + anomalias



Comparativo hidro

13. Limitations

1. Monthly (not daily) data smooths gradual crises
2. Partial ground truth (documented crises, not expert labels)
3. Missing variables (reservoirs, precipitation, spot prices)
4. 85 months Ecuador is minimum viable
5. Consensus does not improve in low-hydro countries (Chile, Argentina)
6. Wide bootstrap CI due to small N [0.40, 1.00]

14. Applications

Energy: CENACE, XM, COES (early warning)
Climate: El Nino impact on hydropower
Academic: multi-technique methodology
Policy: energy diversification

Target journals: Energy Policy, Applied Energy, IEEE Access, Energies (MDPI)

15. References

[1] Liu et al. (2008). Isolation Forest. IEEE ICDM.
[2] Cleveland et al. (1990). STL Decomposition.
[3] Page (1954). CUSUM. Biometrika.
[4] Chandola et al. (2009). Anomaly Detection Survey. ACM.
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